Hackathon report run 2026-11-05 17:37: B_r1_best

Summary

Cost of thermal supply	Investment	Operating cost	CO2 emissions
29.55 EUR/MWh	17 MEUR	2.798 MEUR/yr	11.38 kt

Heat delivered	Cold delivered	Geothermal share of heat	Heat-pump electricity share
111.7 GWh	23.74 GWh	63%	1%

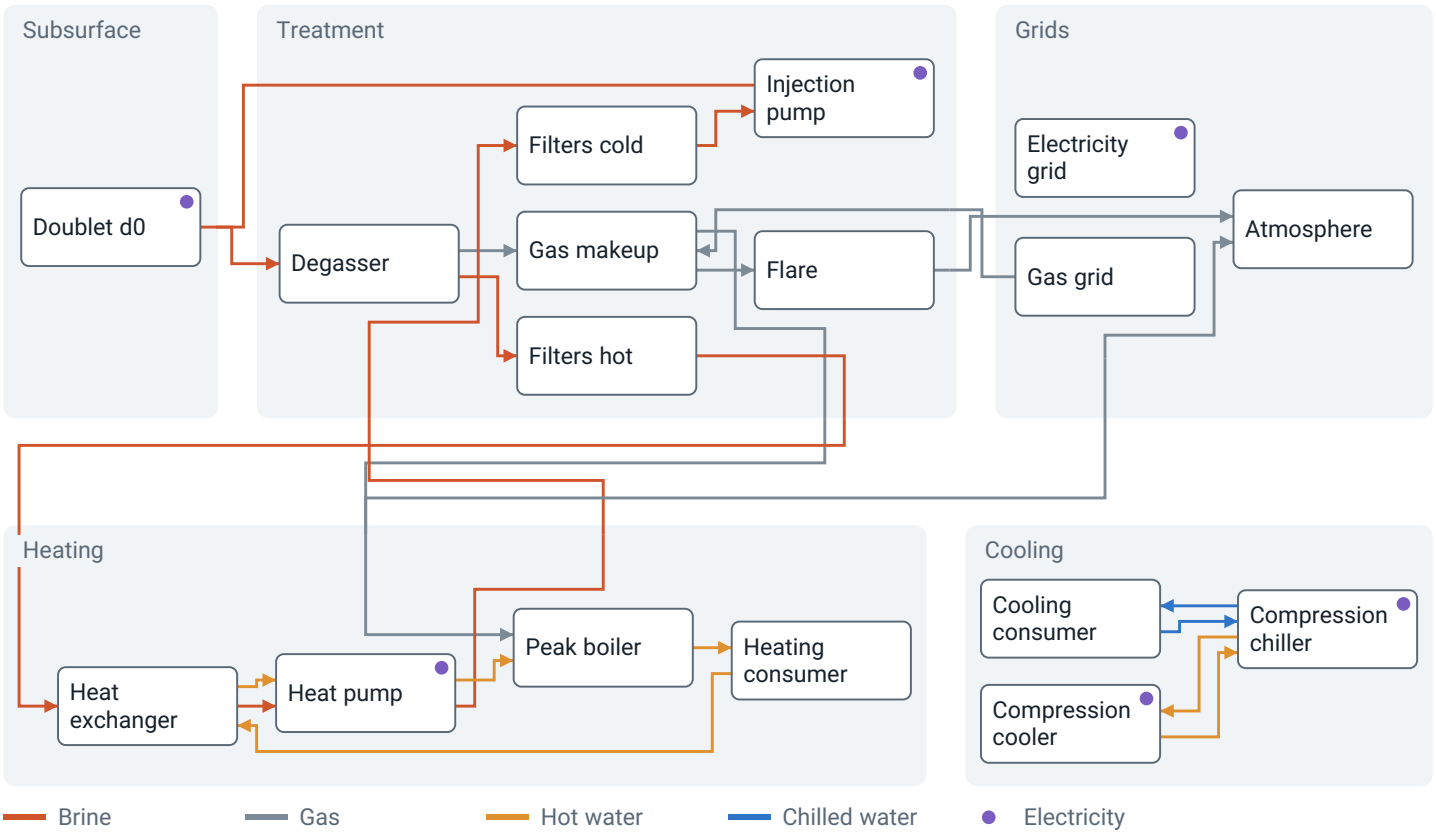
The design delivers 111.7 GWh of heat and 23.74 GWh of cold a year. The levelised cost is 29.55 EUR/MWh. Of the heating demand, geothermal brine heat 63%, heat-pump electricity 1%, backup 36%. The largest rated units are Peak boiler 30 MW, Compression cooler 20 MW, Compression chiller 10 MW. All 12 assessed checks pass.

Plant

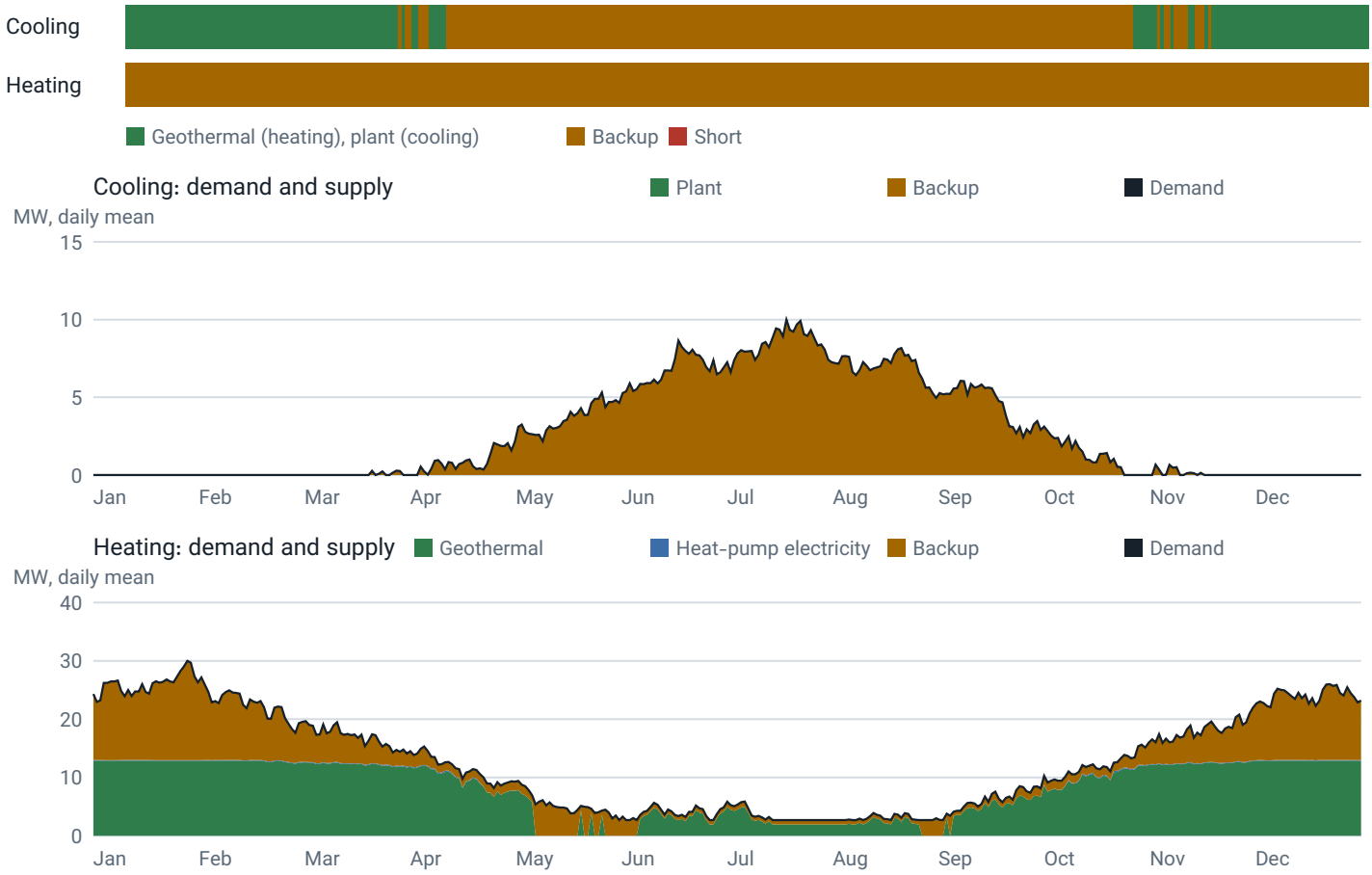
Design B_r1_best from a study of kind Find the best design, plant type B_deep.

Capacities

Module	Kind	Rated	Peak	Used
d0	Production pump	300 m3/h	300 m3/h	100%
Heat pump	Heat pump	2 MW	2 MW	100%
Peak boiler	Boiler	30 MW	17.03 MW	57%
Heat exchanger	Heat exchanger	400 kW/K	-	-
Compression cooler	Cooler, heat rejected	20 MW	11.04 MW	55%
Compression chiller	Compression chiller	10 MW	10 MW	100%



Days covered



Service	Geothermal or plant	Backup	Short
Cooling	146	219	0
Heating	0	365	0

On cooling days the whole supply is the plant: gemini has no cooling backup.

Constraint checks

12 passed, 0 failed, 0 not assessed.

Check	Status	Value	Limit	Margin	Worst day
Doublet count	pass	1	1 to 3	0	-
Cold delivered (Cooling consumer)	pass	0 %	≤ 1 %	1 %	-
Heat delivered (Heating consumer)	pass	0 %	≤ 1 %	1 %	-
Pump efficiency (d0)	pass	70 %	= 70 %	-	-
Full reinjection (d0)	pass	1	1 to 1	0	2027-05-02
Running time limit (d0)	pass	330 d	≤ 330 d	0 d	-
Brine cooling limit (d0)	pass	35.59 K	≤ 40 K	4.411 K	2027-01-01
Production flow range (d0)	pass	300 m3/h	50 to 300 m3/h	0 m3/h	2027-01-01

Check	Status	Value	Limit	Margin	Worst day
Cooling supply temperature (Cooling consumer)	pass	20 degC	= 20 degC	-	2027-03-22
Heating supply temperature (Heating consumer)	pass	100 degC	= 100 degC	-	2027-01-01
Heating supply temperature (service supply rule) (Heating consumer)	pass	100 degC	≥ 100 degC	0 degC	2027-01-01
Cooling supply temperature (service supply rule) (Cooling consumer)	pass	20 degC	≤ 20 degC	0 degC	2027-03-22

Costs

From case version 3, plant version None.

Economics

Figure	Value
Cost of cold	33.71 EUR/MWh
Cost of heat	28.67 EUR/MWh
Cost of thermal supply	29.55 EUR/MWh
Operating cost	2.798 MEUR/yr
Investment	17 MEUR
Annualised cost	4.004 MEUR/yr

Energy balance

Figure	Value
Gas bought	34.06 GWh
Heat from the boilers	40.42 GWh
Cold delivered	23.74 GWh
Heat delivered	111.7 GWh
Electricity bought	5.921 GWh

Emissions

Figure	Value
CO2 emissions	11.38 kt

Other figures

Figure	Value
Cop	18.86
Peak cold demand	10 MW

Figure	Value
Peak heat demand	30 MW
Cooling coverage	1
Heating coverage	1

Capital cost	Amount
Wells	7.049 MEUR
Surface plant	9.599 MEUR
Other	0.35 MEUR

Capital cost by kind

Item	Amount	Unit cost	Source
Drilling	6.815 MEUR	1,239 EUR/m	economics.json, capex.drilling
Compression chiller	3 MEUR	0.3 EUR/W	economics.json, capex.compression_chiller
Cooler	2 MEUR	0.1 EUR/W	economics.json, capex.cooler
Boiler	1.822 MEUR	0.06073 EUR/W	economics.json, capex.boiler
Heat pump	1.476 MEUR	0.7379 EUR/W	economics.json, capex.heat_pump
Degasser	1 MEUR	1,000,000 EUR	economics.json, capex.degasser
Production pump	0.2334 MEUR	0.5 EUR/W	economics.json, capex.production_pump
Heat exchanger	0.2 MEUR	0.5 EUR/(W/K)	economics.json, capex.heat_exchanger
Pipe	0.2 MEUR	500 EUR/m	economics.json, capex.pipe
Injection pump	0.1012 MEUR	0.5 EUR/W	economics.json, capex.injection_pump
Flare	0.1 MEUR	100,000 EUR	economics.json, capex.flare
Gas makeup	0.05 MEUR	50,000 EUR	economics.json, capex.gas_makeup

Uncertainty and sensitivity

This result is not a sensitivity or uncertainty study.

Agent log and tokens

- Status: running
- Input tokens: 9.22 M
- Output tokens: 239 K
- Cost: 0.20 USD

Appendix: modules and connections

Module	Kind	Size	Peak	Used
Cooling return pipe	Pipe	100 m	-	-
Cooling supply pipe	Pipe	100 m	-	-

Module	Kind	Size	Peak	Used
Heating return pipe	Pipe	100 m	-	-
Heating supply pipe	Pipe	100 m	-	-

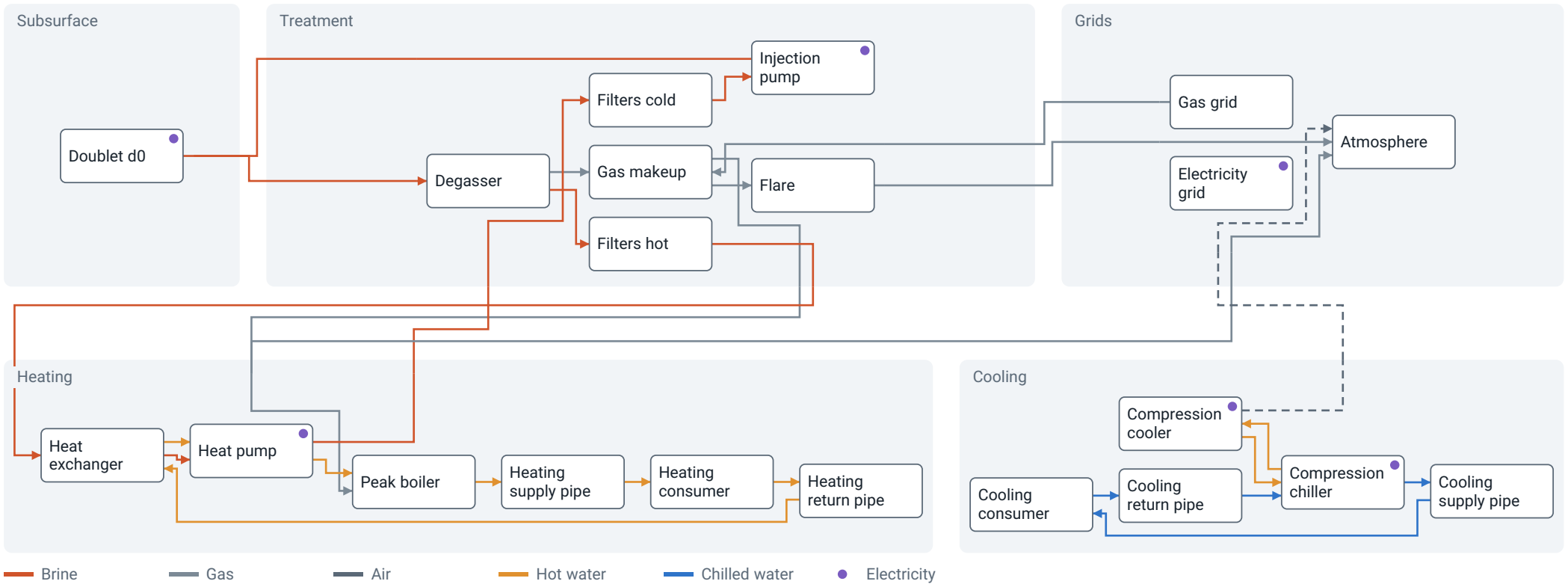
Plant: B_deep

Module	Type	Category	Connected to	Parameters
d0	Doublet	plants	return_in <- injection_pump.fluid_out, power_in <- electricity_grid.power_out	85
flare	Flare	treatment	gas_in <- gas_makeup.excess_out	1
filters.hot	Filter	treatment	fluid_in <- degasser.liquid_out	3
filters.cold	Filter	treatment	fluid_in <- heat_pump.source_out	3
degasser	Degasser	treatment	fluid_in <- d0.supply_out	4
gas_grid	GasGrid	grids		1
heat_pump	HeatPump	heating	source_in <- heat_exchanger.hot_out, sink_in <- heat_exchanger.cold_out, power_in <- electricity_grid.power_out	9
atmosphere	Atmosphere	environment	gas_in <- peak_boiler.flue_out, gas_in <- flare.flue_out, heat_in <- flare.loss_out, air_in <- compression_cooler.air_out, heat_in <- heating_return_pipe.loss_out, heat_in <- heating_supply_pipe.loss_out, heat_in <- cooling_return_pipe.loss_out, heat_in <- cooling_supply_pipe.loss_out, heat_in <- peak_boiler.loss_out	1

Module	Type	Category	Connected to	Parameters
gas_makeup	GasMakeup	piping	gas_in <- degasser.gas_out, grid_in <- gas_grid.gas_out	0
peak_boiler	Boiler	heating	fuel_in <- gas_makeup.fuel_out, fluid_in <- heat_pump.sink_out	3
heat_exchanger	HeatExchanger	exchangers	hot_in <- filters.hot.fluid_out, cold_in <- heating_return_pipe.fluid_out	8
injection_pump	InjectionPump	pumps	fluid_in <- filters.cold.fluid_out, power_in <- electricity_grid.power_out	7
cooling_consumer	CoolingGrid	grids	supply_in <- cooling_supply_pipe.fluid_out	5
electricity_grid	ElectricityGrid	grids		2
heating_consumer	HeatingGrid	grids	supply_in <- heating_supply_pipe.fluid_out	4
compression_cooler	Cooler	exchangers	fluid_in <- compression_chiller.sink_out, air_in <- atmosphere.air_out, power_in <- electricity_grid.power_out	6
compression_chiller	HeatPump	heating	power_in <- electricity_grid.power_out, sink_in <- compression_cooler.fluid_out, source_in <- cooling_return_pipe.fluid_out	9

Module	Type	Category	Connected to	Parameters
cooling_return_pipe	Pipe	piping	fluid_in <- cooling_consumer.return_out, air_in <- atmosphere.air_out	6
cooling_supply_pipe	Pipe	piping	fluid_in <- compression_chiller.source_out, air_in <- atmosphere.air_out	6
heating_return_pipe	Pipe	piping	fluid_in <- heating_consumer.return_out, air_in <- atmosphere.air_out	6
heating_supply_pipe	Pipe	piping	air_in <- atmosphere.air_out, fluid_in <- peak_boiler.fluid_out	6

Total parameters: 178


Subsurface
Treatment
Grids
Doublet d0
Degasser
Injection pump
Filters cold
Gas makeup
Filters hot
Flare
Gas grid
Atmosphere
Electricity grid
Heating
Heat exchanger
Heat pump
Peak boiler
Heating supply pipe
Heating consumer
Heating return pipe
Cooling
Compression cooler
Compression chiller
Cooling consumer
Cooling return pipe
Cooling supply pipe
Brine
Gas
Air
Hot water
Chilled water
Electricity